# Additional file 1

## Supplementary tables and CLAIM checklist

*Segmentation, type classification and measurement of brain tumors on single 2D MRI slices: a retrospective development and leakage-controlled internal validation study*

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**Table S1 Completed CLAIM checklist**

| Item | Topic | Where reported |
|---|---|---|
| 1 | Identification as a study of AI methodology | Title; Abstract |
| 2 | Structured summary | Abstract (Background, Methods, Results, Conclusions) |
| 3 | Scientific and clinical background, intended use | Background |
| 4 | Study objectives and hypotheses | Background, last paragraph |
| 5 | Prospective or retrospective study | Methods: Study design (retrospective) |
| 6 | Study goal (model creation, exploratory, feasibility, non-inferiority) | Methods: Study design (development and internal validation) |
| 7 | Data sources | Methods: Datasets |
| 8 | Eligibility criteria | Methods: Datasets (all slices of both public datasets used) |
| 9 | Data pre-processing | Methods: Preprocessing |
| 10 | Selection of data subsets | Methods: Data splits and leakage control; Table S2 |
| 11 | Definitions of data elements | Methods: Datasets; Additional files 2-4 (column definitions) |
| 12 | De-identification methods | Declarations: Ethics (public de-identified data) |
| 13 | How missing data were handled | Not applicable: no missing images or masks in figshare; Kaggle has no masks by design (Methods: Training) |
| 14 | Definition of ground truth reference standard | Methods: Datasets (expert masks; dataset class labels) |
| 15 | Rationale for choosing the reference standard | Methods: Datasets; Discussion (expert masks and their extent) |
| 16 | Source of ground-truth annotations; qualifications of annotators | Methods: Datasets (annotations by the dataset creators [6-8]; qualifications as reported there) |
| 17 | Annotation tools | Not applicable: masks taken from the public dataset |
| 18 | Measurement of inter- and intrarater variability | Not available for the public masks; stated as limitation of reference standards (Background [4]) |
| 19 | Intended sample size and how it was determined | Methods: Data splits (all available data; no sample-size calculation) |
| 20 | How data were assigned to partitions; proportions | Methods: Data splits and leakage control; Additional file 2 |
| 21 | Level at which partitions are disjoint | Methods: Data splits (patient level for figshare; image-level copy removal for Kaggle) |
| 22 | Detailed description of model | Methods: Model |
| 23 | Software libraries, frameworks and packages | Methods: Training; Table S3; Availability of data and materials |
| 24 | Initialization of model parameters | Methods: Model (ImageNet weights for the encoder) |
| 25 | Details of training approach (augmentation, hyper-parameters, stopping) | Methods: Training; Table S3 |
| 26 | Method of selecting the final model | Methods: Training (best validation score) |
| 27 | Ensembling techniques | Methods: Inference (horizontal-flip averaging only); Table S6 |
| 28 | Metrics of model performance | Methods: Evaluation and statistics |
| 29 | Statistical measures of significance and uncertainty | Methods: Evaluation and statistics (bootstrap 95% CI) |
| 30 | Robustness or sensitivity analysis | Results: Calibration, ablations; Table S6 |
| 31 | Methods for explainability or interpretability | Results: Fig. 2 and Fig. 6 (outlines shown with the prediction) |
| 32 | Validation or testing on external data | Not performed; stated in Limitations |
| 33 | Flow of participants or cases | Methods: Data splits; Table S2 |
| 34 | Demographic and clinical characteristics of cases | Not available in the public datasets; tumor-type composition in Methods: Datasets |
| 35 | Performance metrics for optimal model on all data partitions | Results; Tables 1-3; Fig. 5 (validation curves) |
| 36 | Estimates of diagnostic accuracy and their precision | Results; Table 3 (95% CI) |

| Item | Topic | Where reported |
|---|---|---|
| 37 | Failure analysis of incorrectly classified cases | Results: Fig. 2 (worst cases); Discussion; Additional files 3-4 |
| 38 | Study limitations | Discussion: Limitations |
| 39 | Implications for practice | Discussion; Conclusions |
| 40 | Registration number and name of registry | Not applicable (retrospective study on public data) |
| 41 | Where the full study protocol can be accessed | Methods; source code repository [32] |
| 42 | Sources of funding and other support | Declarations: Funding |

CLAIM, Checklist for Artificial Intelligence in Medical Imaging (Mongan et al., Radiol Artif Intell 2020). Section names refer to the main manuscript; numbers in brackets are its references. Item wording is abbreviated; the official checklist should be consulted for the full text.

## Table S2 Dataset partitions and leakage control

| Partition | Size | Use |
|---|---|---|
| figshare training | 186 patients / 2439 slices | Segmentation and type losses |
| figshare validation | 23 patients / 330 slices | Model selection |
| figshare test | 24 patients / 295 slices | Segmentation, measurement, side and type evaluation |
| Kaggle Training, removed | 305 images | Copies of figshare validation or test slices |
| Kaggle training subset | 4867 images | Classification loss (no-tumor images also segmentation loss) |
| Kaggle validation subset | 540 images | Model selection |
| Kaggle Testing, all | 1311 images | Classification and detection evaluation |
| Kaggle Testing, copies of a training image | 621 images | Reported separately |
| Kaggle Testing, no copies | 690 images (no tumor 28, glioma 243, meningioma 177, pituitary 242) | Leakage-controlled subset |

Copies were identified by comparing 32 × 32 z-scored thumbnails (mean absolute difference < 0.08). The patient-level split is listed in Additional file 2.

## Table S3 Training set-up and software

| Item | Setting |
|---|---|
| Input | 256 × 256 grayscale, zero-padded to square, scaled to [0, 1] |
| Architecture | U-Net decoder, ResNet34 encoder (ImageNet weights), 24.4 million parameters |
| Heads | Segmentation (1 channel, sigmoid); classification (4 classes, dropout 0.3) |
| Loss | BCE + soft Dice (segmentation) + 0.5 × weighted cross-entropy, label smoothing 0.05; class weights 1, 1, 1.2, 1 |
| Optimizer / schedule | AdamW, weight decay 1e-4; one-cycle, maximum learning rate 3e-4, 10% warm-up |
| Batch size / epochs | 12 / up to 12 (selected epoch 11) |
| Data per epoch | All figshare training slices + 1500 random Kaggle training images |
| Augmentation | Horizontal flip p = 0.5; rotation ±15°; scale 0.85–1.15; translation ±8%; gamma 0.7–1.5; contrast ×0.85–1.15; brightness ±0.08; Gaussian blur p = 0.2; Gaussian noise p = 0.3 |
| Selection criterion | Validation Dice (figshare) + validation accuracy (Kaggle) |
| Inference | Average with horizontally flipped input; mask threshold 0.5 |
| Hardware | 12-core CPU, no GPU; about 13 min per epoch |
| Software | Python 3.12.10; PyTorch 2.5.1; torchvision 0.20.1; NumPy 2.3.5; SciPy 1.17.0; scikit-image 0.26.0; OpenCV 4.13.0; scikit-learn 1.8.0 |
| Random seed | Patient split 42; training 0 |

BCE, binary cross-entropy; CPU, central processing unit.

## Table S4 Per-class classification metrics on the Kaggle Testing folder

| Class | Images | Precision | Recall | F1 |
|---|---|---|---|---|
| No tumor | 405 | 0.998 | 0.995 | 0.996 |
| Glioma | 300 | 0.987 | 0.987 | 0.987 |
| Meningioma | 306 | 0.990 | 0.977 | 0.984 |

| Class | Images | Precision | Recall | F1 |
|---|---|---|---|---|
| Pituitary tumor | 300 | 0.980 | 0.997 | 0.988 |

Four-class accuracy 98.9% (95% CI 98.4–99.5%), macro-F1 0.989. Confusion matrix rows (true class: predicted no tumor / glioma / meningioma / pituitary) — No tumor: 403 / 1 / 1 / 0; Glioma: 0 / 296 / 1 / 3; Meningioma: 1 / 3 / 299 / 3; Pituitary tumor: 0 / 0 / 1 / 299.

## Table S5 Measurement, side and scale-estimation details

| Quantity | Result |
|---|---|
| Longest diameter, all outlines (n = 292) | median absolute error 3.6 mm; mean 9.4 mm; bias -0.8 mm; 95% limits -32.2 to 30.6 mm; r = 0.72 |
| Longest diameter, overlapping outlines (n = 269) | median absolute error 3.0 mm; mean 8.6 mm; bias -0.1 mm; 95% limits -30.1 to 29.9 mm; r = 0.75 |
| Tumor area | r = 0.79; median relative error 14.6% |
| Side agreement | 90.4% of 104 eligible slices; 100% of 90 slices with overlapping outline |
| Scale estimate / nominal spacing | ratio 10th, 50th, 90th percentile 0.73, 0.84, 0.98 |
| Low-confidence flag (mean probability < 0.8) | 39 of 618 validation and test outlines missed the tumor; flagged 15 of these and 11 correct outlines |
| Segmentation failures | 26 slices with Dice < 0.1; 3 slices without outline |
| Run time | 0.15 s median (90th percentile 0.17 s) per image on 6 CPU threads |

Diameters were measured on predicted and expert masks with the nominal 0.98 mm pixel size of the 256-pixel figshare images. Limits are Bland–Altman 95% limits of agreement.

## Table S6 Effect of mask threshold and test-time flipping on Dice

| Threshold | Dice mean, no flipping | Dice median, no flipping | Dice mean, flipping | Dice median, flipping |
|---|---|---|---|---|
| 0.3 | 0.756 | 0.870 | 0.755 | 0.860 |
| 0.5 | 0.755 | 0.866 | 0.750 | 0.863 |
| 0.7 | 0.753 | 0.863 | 0.743 | 0.865 |

figshare test patients (24 patients, 295 slices). The manuscript reports the setting with flipping and threshold 0.5.